

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

sns.set(style="whitegrid")
%matplotlib inline
```

```
df = pd.read_csv("Titanic Data.csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Unnamed: 12	Unnamed: 13
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S	NaN	NaN
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C	NaN	NaN
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S	NaN	NaN
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S	NaN	NaN
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S	NaN	NaN

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.shape
```

(1309, 15)

```
df.columns
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
      'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked', 'Unnamed: 12',
      'Unnamed: 13', 'Unnamed: 14'],
      dtype='object')
```

```
df.drop(columns=["Unnamed: 12", "Unnamed: 13", "Unnamed: 14"], inplace=True)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1309 entries, 0 to 1308
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  1309 non-null  int64
1   Survived     1309 non-null  int64
2   Pclass       1309 non-null  int64
3   Name         1309 non-null  object
4   Sex          1309 non-null  object
5   Age          1046 non-null  float64
6   SibSp        1309 non-null  int64
7   Parch        1309 non-null  int64
8   Ticket       1309 non-null  object
9   Fare         1308 non-null  float64
10  Cabin        295 non-null   object
11  Embarked     1307 non-null  object
dtypes: float64(2), int64(5), object(5)
memory usage: 122.8+ KB
```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	
count	1309.000000	1309.000000	1309.000000	1046.000000	1309.000000	1309.000000	1308.000000	
mean	655.000000	0.377387	2.294882	29.881138	0.498854	0.385027	33.295479	
std	378.020061	0.484918	0.837836	14.413493	1.041658	0.865560	51.758668	
min	1.000000	0.000000	1.000000	0.170000	0.000000	0.000000	0.000000	
25%	328.000000	0.000000	2.000000	21.000000	0.000000	0.000000	7.895800	
50%	655.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200	
75%	982.000000	1.000000	3.000000	39.000000	1.000000	0.000000	31.275000	
max	1309.000000	1.000000	3.000000	80.000000	8.000000	9.000000	512.329200	

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	263
SibSp	0
Parch	0
Ticket	0
Fare	1
Cabin	1014
Embarked	2

dtype: int64

```
df["Sex"].value_counts()
```

	count
Sex	
male	843
female	466

dtype: int64

```
df["Pclass"].value_counts()
```

	count
Pclass	
3	709
1	323
2	277

dtype: int64

```
df["Survived"].value_counts()
```

	count
Survived	
0	815
1	494

dtype: int64

```
df["Embarked"].value_counts()
```

	count
Embarked	
S	914
C	270
Q	123

dtype: int64

```
plt.figure(figsize=(6,4))

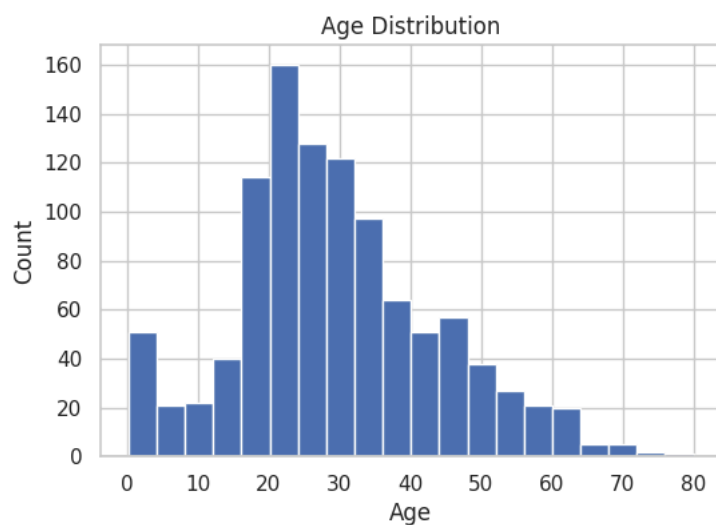
plt.hist(df["Age"].dropna(), bins=20)

plt.title("Age Distribution")

plt.xlabel("Age")

plt.ylabel("Count")

plt.show()
```

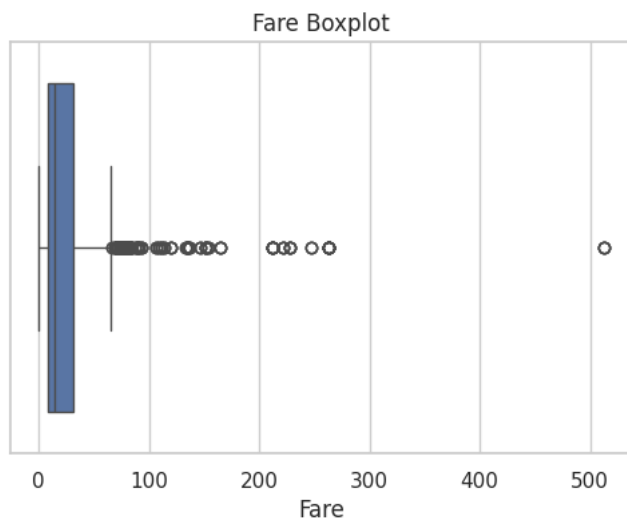


```
plt.figure(figsize=(6,4))

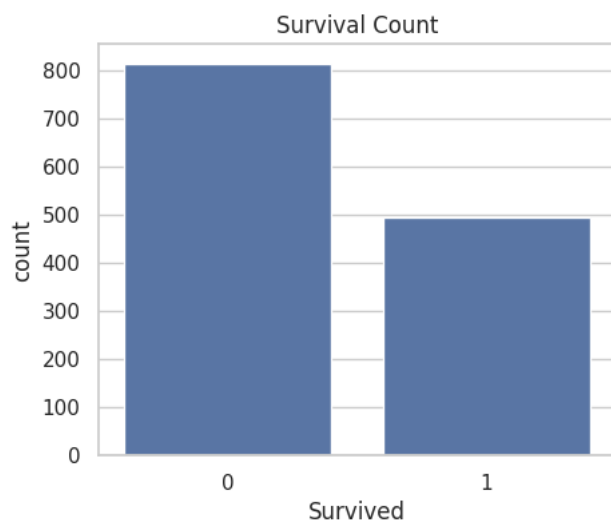
sns.boxplot(x=df["Fare"])

plt.title("Fare Boxplot")

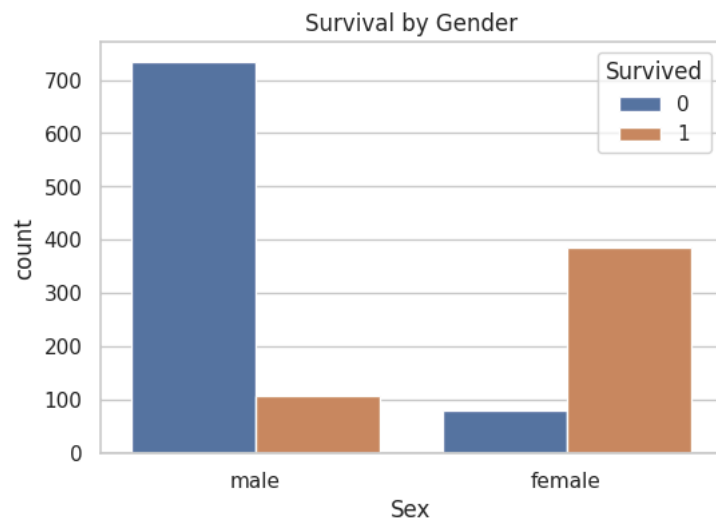
plt.show()
```



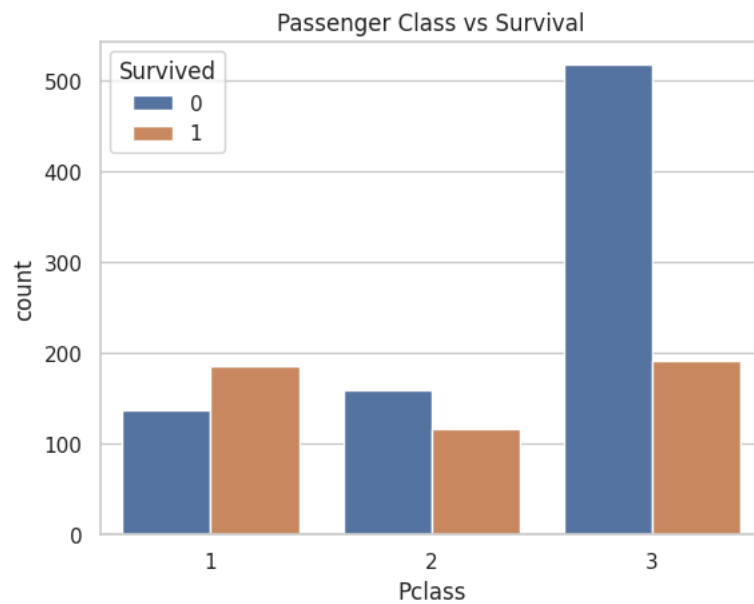
```
plt.figure(figsize=(5,4))  
  
sns.countplot(x="Survived", data=df)  
  
plt.title("Survival Count")  
  
plt.show()
```



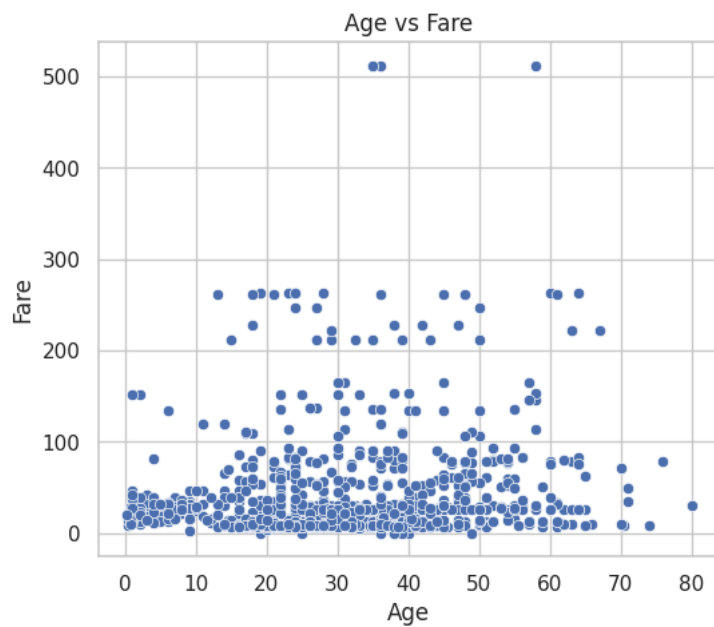
```
plt.figure(figsize=(6,4))  
  
sns.countplot(x="Sex", hue="Survived", data=df)  
  
plt.title("Survival by Gender")  
  
plt.show()
```



```
sns.countplot(x="Pclass", hue="Survived", data=df)  
  
plt.title("Passenger Class vs Survival")  
  
plt.show()
```



```
plt.figure(figsize=(6,5))  
  
sns.scatterplot(x="Age", y="Fare", data=df)  
  
plt.title("Age vs Fare")  
  
plt.show()
```



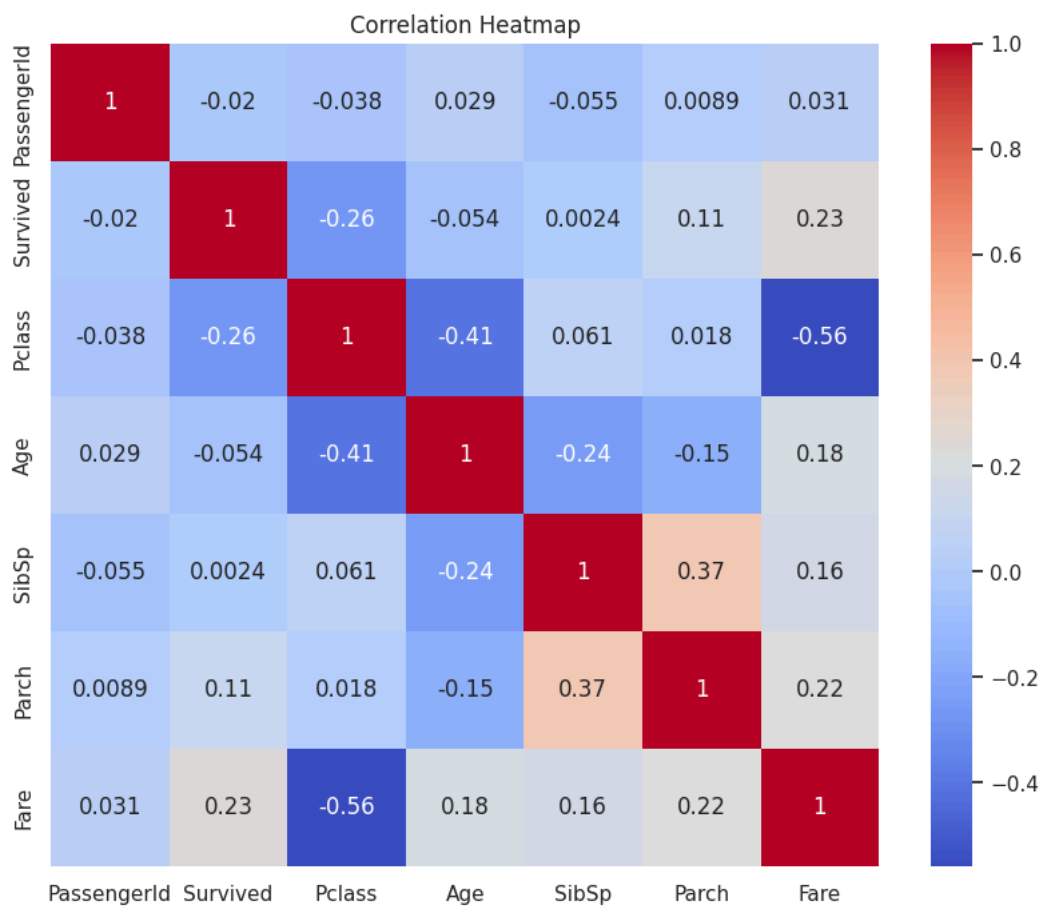
```
numeric_df = df.select_dtypes(include=["number"])

plt.figure(figsize=(10,8))

sns.heatmap(numeric_df.corr(),
            annot=True,
            cmap="coolwarm")

plt.title("Correlation Heatmap")

plt.show()
```



```
sns.pairplot(
    numeric_df[
        ["Age", "Fare", "Pclass", "Survived"]
    ].dropna()
)
```

plt.show()

